Caleb McMullin

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Education

California State University, Fullerton — Fullerton, CA
B.S. in Computer Science — *Aug 2025 – May 2027*

Irvine Valley College — Irvine, CA
Completed lower-division coursework in Computer Science before transferring — *Aug 2023 – May 2025*

Relevant Coursework
Software Engineering, Algorithm Engineering, Cybersecurity

Technical Skills

Languages: Python, JavaScript, SQL, C/C++, C#, HTML/CSS
Frameworks/Libraries: FastAPI, Flask, React, Node.js, pytest, Pydantic, aiosqlite, Vite
Cloud/DevOps: Docker, Docker Compose, GitHub Actions, GHCR, Linux, Ubuntu Server, Tailscale, CI/CD
Databases/Tools: SQLite, MongoDB, Git, GitHub, Jira, Arduino, REST, OAuth 2.0, RBAC

Projects

Garage Sentinel | *Python, FastAPI, Docker, Linux, GitHub Actions, Tailscale, pytest*

- Architected and maintain a 24/7 self-hosted **cloud storage platform** on a dedicated Ubuntu Linux server, managing physical infrastructure, persistent storage, networking, and containerized production services.
- Engineered a fully automated **CI/CD** pipeline using GitHub Actions, Docker Compose, and GHCR, deploying versioned container releases in **1–3 minutes** with automated health verification.
- Developed asynchronous **FastAPI** REST services for media storage and system telemetry, implementing multi-provider OAuth, role-based authorization, secure account linking, and automated pytest tests.
- Configured private remote access via **Tailscale**, enabling secure connectivity to the production server from outside the local network without publicly exposing application ports.

IoT DJ Lighting System | *Python, Flask, JavaScript, C++, Arduino, UDP*

- Designed and fabricated **5–8 custom IoT lighting tubes** using ESP8266 microcontrollers, 60-LED arrays, and 3D-printed enclosures for **$12 per unit**, achieving an estimated **88% reduction in hardware costs** versus commercial alternatives.
- Engineered a distributed Wi-Fi control system using custom **UDP** communication and embedded C++ firmware, supporting automatic device discovery, connectivity recovery, and synchronized animations.
- Built a **Flask-based control interface** enabling real-time RGB lighting, BPM-synchronized animations, and independent device control from anywhere within local Wi-Fi coverage; tested through extended operation and live events.

Tuffy's Belly | *Python, Flask, React, MongoDB, Jira*

- Led a **5-member Agile development team**, coordinating Jira sprints, task assignments, and GitHub code reviews to deliver a full-stack campus dining platform.
- Developed a **React and Flask REST API** application covering **15 dining locations**, integrating interactive Leaflet maps, MongoDB rating aggregation, and authenticated review management.

McMullin Estates | *Python, FastAPI, React, OAuth 2.0*

- Built a **full-stack** real estate application integrating a **React** frontend and FastAPI REST backend with **Google OAuth** 2.0, session authentication, and responsive client-side navigation.

Experience

Uber Driver — May 2018 – June 2023
Uber — *Irvine, CA*

- Completed 4,700–7,200 deliveries a year while maintaining a perfect customer rating, applying strong time-management to ensure consistently on-time deliveries.

Extracurricular

Association for Computing Machinery (ACM) | *Club Member* — Aug 2025 – Present